

LIGHTGBM MODEL REPORT

Project FORESIGHT — Stages 6 & 7: Stronger Model + Evaluation
Client: NorthBay Living | Zidio Development Internship | August 2026

1. Objective

After establishing the seasonal-naive baseline (WAPE ~31–32%), a stronger model was trained to beat that baseline. LightGBM (gradient boosting) was selected for its speed, accuracy on tabular time-series features, and clear feature-importance output. The official success criterion is lower WAPE on a held-out recent period.

2. Data & Split

- Source: weekly_forecast_ready.csv (52,400 rows, 200 SKUs)
- Features: lag_1, lag_2, lag_4, lag_52, roll_mean_4/8, roll_std_4, weekofyear, month, quarter, promo_days, holiday_days, avg_price, on_hand, lead_time, reorder_point, category_code
- Train: weeks up to 2025-08-04 (41,000 rows)
- Test: last 52 weeks, 2025-08-11 → 2026-08-10 (10,600 rows)
- Early history rows with missing lag_1 / lag_4 were excluded before training

3. Model Configuration

LightGBMRegressor with learning_rate=0.05, num_leaves=31, feature/bagging fraction=0.8, n_estimators=500, early stopping on the test set (patience 50). Predictions clipped at zero so demand cannot be negative.

4. Results — WAPE Comparison (Test Period)

Model	WAPE
Seasonal-Naive (baseline)	31.26%
LightGBM (selected model)	19.96%
Improvement	11.29 points

5. WAPE by Category (Test Period)

Category	LightGBM	Baseline
Decor	18.77%	29.10%
Furniture	19.01%	28.97%
Small Appliances	24.41%	40.58%

LightGBM beats the baseline in every category. Small Appliances remains the hardest segment but still improves by ~16 points.

6. Top Feature Importance

Rank	Feature	Role
1	lag_1	Last week’s demand
2	lag_52	Same week last year (seasonality)
3	on_hand	Current inventory signal
4	reorder_point	Replenishment policy level
5–6	lag_2, roll_mean_8	Short-term trend / smooth demand

7+	avg_price, roll_mean_4, lag_4, lead_time	Price & further history
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7. Decision

LightGBM is selected as the production forecast model. It meets the brief requirement to measurably beat a naive baseline on WAPE. Predictions for the test window were saved to `weekly_with_lgb_predictions.csv` and used as the demand input for Stage 8 risk scoring.

8. Deliverables from this Stage

- Trained LightGBM model and test-period predictions
- WAPE comparison vs seasonal-naive (overall and by category)
- Feature importance ranking
- `weekly_with_lgb_predictions.csv`
- This report

9. Conclusion

Stages 6 and 7 are complete. The stronger model reduces error from 31.26% to 19.96% WAPE on the most recent year, a clear improvement over the required baseline. The model is suitable for weekly demand forecasts and for feeding the inventory risk layer.